Cover letter (revised submission) to the Editors of *Quantum*

Hikaru Wakaura, Taiki Tanimae
QIRI (Quantum Integrated Research Institute Inc.)
1–16–3, Akasaka, Minato-ku, Tokyo 107–0061, Japan
h.wakaura@qiri.co.jp ORCID: 0009-0001-2341-7710

September 14, 2026

Dear Editors,

On September 14, 2026, we submitted the revised manuscript "**Blind Catalytic Quantum Error Correction: Target-State Estimation and Fidelity Recovery Without A Priori Knowledge**", authored by Hikaru Wakaura and Taiki Tanimae, for consideration as an Article in *Quantum*. The submission consists of the main manuscript (27 pages, 17 figures, 8 tables, single bibliography), a point-to-point response to the two reviewers, and a frozen release of the open-source package `blind_cqec` v0.2.0 (MIT license, https://github.com/deeptell-inc/blind_cqec_pkg) with a 91-test reproducibility suite, shipped raw results files, and a script-to-figure map.

What we discovered. Catalytic quantum error correction (CQEC) is a threshold-free recovery framework whose original formulation requires the ideal target state to be known exactly. We discovered that the target-knowledge requirement can be displaced onto a classical estimation problem: under an explicitly stated idealized recovery interface—the mode-inclusion-restricted projection of the estimate—recovery fidelity is bounded by estimation fidelity plus a mode-survival penalty, and this bound is realized in both of its regimes across an 84-condition benchmark. The resulting design-space map is sharp: with exactly calibrated noise, channel inversion coincides with the oracle at every dimension and both are capped by a mode-survival ceiling; the noise-model-free alternative, coherence maximization, matches the ceiling to within 1.5% at $d \leq 16$ and degrades with dimension; and the operational choice between the two is decided by calibration accuracy, with the dephasing rate exponentially dominant.

Why this matters. The result separates, for the first time in the CQEC setting, what is a classical estimation question from what is a structural limit of catalytic recovery, and it does so with the negative results reported alongside the positive ones: coherence maximization is actively harmful for mixed targets, no strategy reaches high fidelity where mode annihilation sets the ceiling, and universal single-copy recovery is impossible for any deterministic target-independent apparatus. We believe this honest map of where blind recovery can and cannot work is the prerequisite for the two open problems the manuscript identifies—an explicit joint covariant recovery channel and an operational measurement model—which we state as future work rather than claim.

Why *Quantum*. The previous round of review at *Quantum* was exceptionally rigorous, and the revised manuscript is substantially the product of it: the recovery interface is now stated once and declared as unsimulated physics, the catalytic theorem is restated correctly, a general Lipschitz lemma is replaced by the rigorous one-sided projection bound that a reviewer derived, an algebraically inconsistent crossover formula is removed, every "single-copy" and tomographic-saturation claim is withdrawn, and all numerics are regenerated from a single forward/inverse-consistent noise implementation with raw data released. We believe the resulting paper—narrower in claim, fully reproducible, and explicit about its scope—is well matched to *Quantum*'s standards of rigor and open science, and we are grateful to the reviewers for the direction.

Suggested referees. We would be glad for the manuscript to return to the original reviewers. Additional suggestions, should they be needed:

1. **Ryuji Takagi**, Nanyang Technological University, Singapore (ryuji.takagi@ntu.edu.sg). Co-author of the foundational work on catalytic coherence transformations.

2. **Mark M. Wilde**, Cornell University, USA (wilde@cornell.edu). Quantum Shannon theory, channel fidelity bounds.

3. **Andreas Winter**, Universitat Autònoma de Barcelona, Spain (andreas.winter@uab.cat). Catalytic resource theories.

4. **Christophe Piveteau**, IBM Research Zürich, Switzerland (cpv@zurich.ibm.com). Probabilistic error cancellation.

5. **Bálint Koczor**, University of Oxford, UK (balint.koczor@maths.ox.ac.uk). Error mitigation, near-term VQE.

6. **Daniel Stilck França**, ENS Lyon / CNRS, France (daniel.stilck-franca@ens-lyon.fr). Quantum state estimation, copy complexity.

7. **Tim Byrnes**, NYU Shanghai, China (tim.byrnes@nyu.edu). Co-author of the purification-based suppression work cited in our comparison.

8. **Martí Perarnau-Llobet**, University of Geneva, Switzerland (marti.perarnaullobet@unige.ch). Catalysis and asymmetry resource theory.

9. **Iman Marvian**, Duke University, USA (iman.marvian@duke.edu). Asymmetry resource theory, covariant operations.

10. **Patrick Hayden**, Stanford University, USA (phayden@stanford.edu). Fidelity-based recovery and decoupling.

Non-preferred referees. None.

Corresponding author. Hikaru Wakaura, QIRI Inc., 1–16–3, Akasaka, Minato-ku, Tokyo 107–0061, Japan; h.wakaura@qiri.co.jp; ORCID 0009-0001-2341-7710. Co-author Taiki Tanimae (t.tanimae@qiri.co.jp) is at the same affiliation.

Statements. This work has not been published or submitted elsewhere. Both authors have read and approved the submission. We declare no competing interests.

We thank you for your consideration and look forward to the referees' feedback.

Sincerely,

Hikaru Wakaura
Taiki Tanimae
QIRI Inc., Tokyo